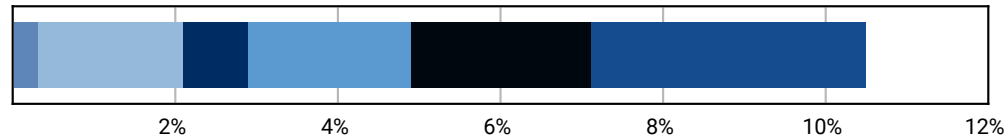


ADV: Statistics (Adv), S2 Interpretation and Bivariate Data (Adv) Classifying Data (Y12)

Teacher: Denis Vlismas

Exam Equivalent Time: 10.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

S2 Interpretation and Bivariate Data



*Analytics based on the average contribution to the 2ADV/STD2 HSC exams over the past decade.

- Classifying Data
- Bar Charts and Histograms
- Other Chart Types
- Summary Statistics - Box Plots
- Summary Statistics - No Graph
- Bivariate Data Analysis

HISTORICAL CONTRIBUTION

- S2 Interpretation and Bivariate Data* is a Year 12 topic that didn't previously exist in the Mathematics/Advanced course, although it has a decade-long history in the Std2/Gen2 exam.
- It provides an area where examiners can test common content between the *Advanced* and *Standard 2* courses. The above bar chart shows the relative importance of the subtopics in past Std2/Gen2 exams.
- S2 Interpretation and Bivariate Data* has been split into six sub-topics for analysis which are listed in the bar chart above.
- This analysis looks at the smallest sub-topic, *Classifying Data* (0.3%).

HSC ANALYSIS - What to expect and common pitfalls

- Classifying Data* looks at categorical data (nominal vs ordinal) and numerical data (discrete vs continuous).
- Students must be confident in identifying any given variables within the definitions above.
- It has been asked in multiple choice questions 3 times since 2016 (in Std2 exams). We note that questions on stratified sampling are beyond the new *Advanced* syllabus and are not included in the database.

Questions

1. Statistics, STD2 S1 2009 HSC 3 MC

The eye colours of a sample of children were recorded.

When analysing this data, which of the following could be found?

- (A) Mean
- (B) Median
- (C) Mode
- (D) Range

2. Statistics, STD2 S1 SM-Bank 1 MC

A survey asked the following question for students born in Australia:

"Which State or Territory were you born in?"

How would the responses be classified?

- A. Categorical, ordinal
- B. Categorical, nominal
- C. Numerical, discrete
- D. Numerical, continuous

3. Statistics, STD2 S1 2015 HSC 4 MC

On a school report, a student's record of completing homework is graded using the following codes.

C = consistently
U = usually
S = sometimes
R = rarely
N = never

What type of data is this?

- (A) Categorical, ordinal
- (B) Categorical, nominal
- (C) Numerical, continuous
- (D) Numerical, discrete

4. Statistics, STD2 S1 2018 HSC 3 MC

A survey asked the following question.

'How many brothers do you have?'

How would the responses be classified?

- A. Categorical, ordinal
- B. Categorical, nominal
- C. Numerical, discrete
- D. Numerical, continuous

5. Statistics, STD2 S1 2016 HSC 7 MC

Which set of data is classified as categorical and nominal?

- (A) blue, green, yellow
- (B) small, medium, large
- (C) 5.2 cm, 6 cm, 7.21 cm
- (D) 4 people, 5 people, 9 people

6. Statistics, STD2 S1 2011 HSC 25a*

A study on the mobile phone usage of NSW high school students is to be conducted.

Data is to be gathered using a questionnaire.

The questionnaire begins with the three questions shown.

Q1: Do you own a mobile phone?
Yes ☐ No ☐

Q2: Which phone company do you use?
.....

Q3: Do you use pre-paid or a plan?
Pre-paid ☐ Plan ☐

- i. Classify the type of data that will be collected in Q2 of the questionnaire. (1 mark)
- ii. Write a suitable question for this questionnaire that would provide discrete ordinal data. (1 mark)

Worked Solutions

1. Statistics, STD2 S1 2009 HSC 3 MC

Eye colour is categorical data

∴ Only the mode can be found

⇒ C

2. Statistics, STD2 S1 SM-Bank 1 MC

The data is categorical (not numerical) since the name of a State is required.

This data cannot be ordered.

⇒ B

3. Statistics, STD2 S1 2015 HSC 4 MC

The data has been grouped into categories and because each category can be ranked, it is ordinal.

⇒ A

4. Statistics, STD2 S1 2018 HSC 3 MC

The number of brothers a person has is an exact whole number.

∴ Classification is numerical, discrete.

⇒ C

5. Statistics, STD2 S1 2016 HSC 7 MC

Categorical and nominal data is qualitative and not ordered.

⇒ A

♦♦ Mean mark 26%.

6. Statistics, STD2 S1 2011 HSC 25a*

i. Categorical, nominal

**ii. How many outgoing calls do you make per day?
(Ensure it can be answered with a numerical score.)**